Nickolas Simons

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Education

University of Utah — Salt Lake City, UT
Master of Entertainment Arts and Engineering, Engineering — *Aug. 2026 – May 2028*

University of Illinois at Urbana-Champaign — Urbana, IL
Bachelor of Science, Computer Science — *Aug. 2022 – May 2025*
Minor in Game Studies and Design — *GPA: 3.96*

Experience

University of Utah, Division of Games — Salt Lake City, UT
Research Assistant - Digital Twins Research — *Aug. 2026 – Dec. 2026*

University of Illinois, stu/dio — Urbana, IL
Programming Lead — *Aug. 2024 – Dec. 2025*

- Managed a team of 12 programmers across 8 projects, conducting code reviews and providing planning feedback to ensure code quality and maintainability
- Maintained Perforce infrastructure and integrated Horde for automated builds and Helix Swarm for code review
- Established coding standards and development workflows to improve code maintainability across projects

Gameplay Programmer — *April 2024 – Dec. 2025*

- Designed and implemented gameplay features and tools for VR platforms using C++ and Blueprints, with an emphasis on accessibility, performance, and usability in classroom environments
- Optimized gameplay code and materials to meet VR performance targets using profiling tools like Unreal Insights and Nvidia Nsight

Projects

Recycool | *UE5, C++, Blueprint, Perforce* — Sept 2026

- Built a serious VR game for standalone Meta Quest deployment in four weeks for the purpose of teaching children how to properly recycle, incorporating street art and trick shot mechanics for greater user engagement
- Implemented an aim-assist system using C++ and Blueprint to make throwing and physics based interactions more intuitive and fun for users
- Worked in a team of 6, developing tools, design interfaces, and drafting technical documentation for more efficient on-boarding and team communication

VRchaeology | *UE5, C++, Blueprint, Perforce* — May 2025 – Dec 2025

- Designed and implemented a modular quest system with editor tooling using C++ to guide user actions through reusable scripting logic and contextual prompts
- Developed an interactive, tool-based VR excavation system using C++, simulating surface digging and stratigraphic layer analysis
- Implemented shared sublevel instances in UE5 to maintain continuity across multiple uses of a shared environment, ensuring consistent gameplay and scene behavior, and simplifying the workflow for content teams

Multiplayer Action Game | *UE5, C++, Blueprint, ASIO, C#, GraphQL, MongoDB* — Nov. 2025 – April 2026

- Built a multiplayer combat framework using C++ and Epic's Gameplay Ability System, implementing client-side prediction and correction for movement abilities
- Developed a utility service using C#, MongoDB, and GraphQL to integrate with matchmaking, allowing queries for game state information prior to joining a session
- Created a system for generating replicated procedural environments using Epic's new Procedural Content Generation framework

Technical Skills

Proficiencies: Gameplay Engineering, DevOps, Networking, Engine Development
Languages: C++, C, C#, SQL, Python, Haskell, UE Blueprint
Tools: Unreal Engine 5, Perforce, Git, Visual Studio, Unity 6, Docker, Horde, Jenkins
Technologies: SDL3, OpenGL, Dear ImGui, CMake, ASIO (C++), Hot Chocolate (GraphQL), Linux